

OPLOGS-03: OpenPipeline Processing

Series: OPLOGS — OpenPipeline Logs | **Notebook:** 3 of 8 | **Created:** December 2025 | **Last Updated:** 09/28/2026

Configuring Pipeline Stages for Log Transformation

This notebook covers OpenPipeline processing stages: parsing, enrichment, metric extraction, event generation, bucket routing, and filtering.

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Prerequisites

-  Access to a Dynatrace environment with log data
-  OpenPipeline configuration permissions
-  Completed OPLOGS-01 and OPLOGS-02

1. Enrichment at the Source

OneAgent Attribute Enrichment (OneAgent 1.333+): OneAgent can enrich all telemetry (metrics, spans, logs, events) with primary fields (`dt.security_context` , `dt.cost.costcenter`) and primary tags (`primary_tags.environment` , `primary_tags.team`) at the source — feeding directly into OpenPipeline routing, bucket assignment, and Grail permissions. Set them at install time with `--set-host-tag` , on existing hosts

with `oneagentctl --set-host-tag`, or per process with `DT_TAGS`. See [Primary Grail fields and tags enrichment through OneAgent \(DT docs\)](#).

2. Parsing & Field Extraction

Parsing extracts structured fields from unstructured log content **at ingestion time**.

DPL (Dynatrace Pattern Language) Matchers

DPL Pattern Matchers Quick Reference		
Dynatrace Pattern Language for Log Parsing		
Text Matchers		
LD	Line data (minimal)	"error in module"
WORD	Single word	"ERROR"
DATA	Any text, across lines	"any content"
NSPACE	Non-whitespace	"user@host.com"
SPACE	Whitespace characters	" " (tabs/spaces)
Number Matchers		
INT	Integer	-1, 42, 1000
LONG	Long integer	9223372036854775807
DOUBLE	Decimal number	3.14, -0.5
BOOLEAN	true/false	true, false
HEXINT	Hex integer	0x1F, 0xFF
Network / Structured		
IPADDR	IPv4/IPv6 address	10.0.0.1, ::1
IPv4	IPv4 only	192.168.1.100
IPv6	IPv6 only	2001:db8::1
DQS	Double-quoted string	"quoted str"
JSON	JSON object	{"k": "v"}
Time / Date		
TIMESTAMP	Date/time	2024-01-15 14:30:00
ISO8601	ISO 8601 timestamp	2024-01-15T14:30:00Z
TIME	Time — TIME('HH:mm:ss')	14:30:00
DURATION	Time duration	2h30m, 1d
Syntax: <code>MATCHER:fieldname</code> (e.g., <code>INT:status_code</code>) or <code>MATCHER</code> (no capture) · Full catalog: FAQ-15 §4		

Matcher	Description	Example Match
LD	Line data (to delimiter)	Any text
INT	Integer	42 , -17
DOUBLE	Decimal number	3.14 , -0.5
IPADDR	IP address	192.168.1.1
WORD	Word characters	hello123
TIMESTAMP	Date/time	Various formats
JSON	JSON object/array	{"key": "value"}

OpenPipeline Parse Processor

```
# Parse HTTP access logs
processors:
- name: parse-http-logs
  type: dql
  source: content
  dql: |
    parse content, "IPADDR:client_ip SPACE LD SPACE '[' TIMESTAMP:request_time ']'
      SPACE '\"' LD:method SPACE LD:path SPACE LD '\"'
      SPACE INT:status SPACE INT:bytes"
```

```
// Discover log patterns for parsing design
fetch logs, from: now() - 1h
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 20
```

```
// Test parse pattern before configuring in OpenPipeline
fetch logs, from: now() - 1h
| filter contains(content, "GET") OR contains(content, "POST")
| parse content, "LD:method SPACE '/' LD:path SPACE INT:status_code"
| filter isNotNull(status_code)
| summarize {count = count()}, by: {method, status_code}
| sort count desc
| limit 15
```

```
// Verify fields already extracted by OpenPipeline
fetch logs, from: now() - 1h
| summarize {
  has_loglevel = countIf(isNotNull(loglevel)),
  has_status = countIf(isNotNull(status)),
  has_trace_id = countIf(isNotNull(trace_id)),
  has_span_id = countIf(isNotNull(span_id))
}
```

3. Metric Extraction from Logs

Extract metrics with dimensions from log data at ingestion time. This is powerful for:

- Creating SLIs from log patterns
- Building dashboards without log queries
- Enabling metric-based alerting

OpenPipeline Metric Extraction

```
# Extract request duration metric from logs
processors:
  - name: extract-request-metric
    type: metric
    enabled: true
    condition: matchesValue(content, "**duration=")
    metricKey: log.request.duration
    dimensions:
      - service: k8s.namespace.name
      - method: extracted_method
      - status: extracted_status
    value: extracted_duration_ms
```

Write matching conditions without `contains()` . OpenPipeline matchers accept only a subset of DQL, and `contains()` and `in()` are rejected there (the verifier error `The function contains() isn't enabled.`). Use `matchesValue(field, "**text*")` (case-insensitive substring) or `matchesPhrase(field, "text")` (whole words). The conditions in this notebook's configuration sketches are written that way. `contains()` is still fine inside a DQL processor statement and in queries.

Common Metric Extraction Patterns

Log Pattern	Metric	Dimensions
Request completed	<code>log.request.count</code>	service, method, status
Response time	<code>log.response.duration</code>	endpoint, status_code
Error occurred	<code>log.error.count</code>	error_type, service
Queue depth	<code>log.queue.size</code>	queue_name

SaaS 1.337 (April 2026): Recommended Fields for Extraction Processors

When you configure an extraction processor (metric, Davis event, business event or SDLC event), OpenPipeline suggests the fields to extract. Per the SaaS 1.337 release note: *"Recommendations automatically cover all permission- and cost-relevant fields or dimensions, with support for all Smartscape identifiers and Grail primary tags."* — so dimensions such as `dt.security_context` , `dt.cost.costcenter` , `dt.smartscape.*` and `primary_tags.*` are offered by default. Existing processors are unchanged.

Sources: [SaaS 1.337 \(DT docs\)](#)

```
// Identify logs with numeric values for metric extraction
fetch logs, from: now() - 1h
| filter contains(content, "duration")
      OR contains(content, "latency")
      OR contains(content, "time=")
      OR contains(content, "ms")
| fieldsAdd content_preview = substring(content, from: 0, to: 120)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 15
```

```
// Simulate metric extraction: count by dimensions
// status == "ERROR" covers every error-class level (ERROR, SEVERE, CRITICAL, FATAL, ...);
// loglevel == "ERROR" alone misses SEVERE and the rest
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {
  request_count = count(),
  error_count = countIf(status == "ERROR")
}, by: {k8s.namespace.name, k8s.workload.name}
| fieldsAdd error_rate = round((error_count * 100.0) / request_count, decimals: 2)
| sort error_count desc
| limit 15
```

```
// Preview: What dimensions would be valuable?
fetch logs, from: now() - 1h
| summarize {
  unique_namespaces = countDistinct(k8s.namespace.name),
  unique_workloads = countDistinct(k8s.workload.name),
  unique_pods = countDistinct(k8s.pod.name),
  unique_hosts = countDistinct(dt.entity.host),
  unique_sources = countDistinct(dt.openpipeline.source)
}
```

4. Attribute Creation & Enrichment

Sprint 1.345 (August 2026): Inline Lookup Processor

Enrichment that previously needed a DQL `if()` ladder — or an external system — can now be expressed as a **lookup table defined inside the processor**. The **Inline lookup** processor maps an existing attribute to a new or updated value with no external connection.

Fits	Example
Code → description	HTTP or application error code to a human-readable meaning
Identifier → ownership	App ID to business unit or cost centre
Attribute → security context	Existing attribute to a <code>dt.security_context</code> value

It is available in the **Processing** stage across every scope — logs, spans, metrics, business events, security events, user events and sessions, and the Davis/SDLC event pipelines — not logs alone. SaaS 1.345 released 08/11/2026 with a **staged tenant rollout**; verify the processor is present in your pipeline editor before designing

against it. The `if()` -ladder pattern shown below remains correct and keeps working, and is still the better fit when the mapping is computed rather than enumerated.

Add **computed attributes** to logs for enhanced analysis and filtering.

OpenPipeline Field Processor

```
# Add computed attributes
processors:
- name: enrich-environment
  type: fieldsAdd
  fields:
    - name: environment
      value: |
        if(contains(k8s.namespace.name, "prod"), "production",
        else: if(contains(k8s.namespace.name, "staging"), "staging",
        else: "development"))

    - name: severity_score
      value: |
        if(loglevel == "ERROR", 3,
        else: if(loglevel == "WARN", 2,
        else: 1))

    - name: team_owner
      value: |
        if(contains(k8s.namespace.name, "payment"), "platform-team",
        else: if(contains(k8s.namespace.name, "frontend"), "web-team",
        else: "unknown"))
```

Common Attribute Patterns

Attribute	Source	Use Case
environment	namespace name	Filter prod vs dev
team_owner	namespace/labels	Route alerts
severity_score	loglevel	Prioritization
log_category	content patterns	Classification

```
// Preview attribute creation logic
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd environment = if(contains(k8s.namespace.name, "prod"), "production",
                             else: if(contains(k8s.namespace.name, "staging"), "staging",
                             else: "development"))
| fieldsAdd severity_score = if(loglevel == "ERROR", 3,
                              else: if(loglevel == "WARN", 2,
                              else: 1))
| summarize {count = count()}, by: {environment, severity_score, loglevel}
| sort environment asc, severity_score desc
```

```
// Categorize logs by content patterns
fetch logs, from: now() - 1h
| fieldsAdd log_category = if(contains(content, "Exception") OR contains(content, "Error"),
"exception",
                                else: if(contains(content, "request") OR contains(content, "response"),
"http",
                                else: if(contains(content, "database") OR contains(content, "query"),
"database",
                                else: if(contains(content, "auth") OR contains(content, "login"),
"security",
                                else: "general"))))
| summarize {count = count()}, by: {log_category, loglevel}
| sort count desc
```

```
// Identify namespaces for team ownership mapping
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize {log_count = count()}, by: {k8s.namespace.name}
| sort log_count desc
| limit 20
```

5. Event Generation from Logs

Create **business events** from specific log patterns. Events flow to Grail and can trigger workflows.

Business event processor (Data extraction stage)

Event generation is a **Business event** processor in the pipeline's **Data extraction** stage — there is no configuration file to write. In **Settings > Process and contextualize > OpenPipeline > Logs > Pipelines**, open the pipeline, go to **Data extraction**, and add one processor per event. Each takes a *Name*, a *Matching condition*, an *Event type* and an *Event provider* (each either a record field or a static string), and a *Field extraction* choice (all fields, only listed fields, or all except listed fields):

Name	Matching condition	Event type (static)	Event provider (static)	Field extraction
payment-completed	<code>matchesValue(content, "**payment*") AND matchesValue(content, "**completed*")</code>	<code>com.example.payment.completed</code>	payment-service	Fields to extract: payment_id , amount , currency , customer_id
critical-error	<code>status == "ERROR" AND matchesValue(content, "**critical*")</code>	<code>com.example.critical.error</code>	app-logs	Fields to extract: error_type , k8s.namespace.name

The fields you extract must already exist on the record when the Data extraction stage runs — parse `payment_id` , `amount` and the others in the **Processing** stage first (§2). Matching conditions accept only a subset of DQL: `contains()` and `in()` are rejected (the verifier error `The function contains() isn't enabled.`), so substring tests use `matchesValue(field, "**text*")` , which is case-insensitive. Both conditions above verify as valid logs matchers; the error condition uses `status == "ERROR"` so SEVERE, CRITICAL and FATAL records are included (see the error-count query in §3).

Event Use Cases

Log Pattern	Event Type	Processor	Purpose
Order completed	<code>order.completed</code>	Business event	Business analytics
User signup	<code>user.registered</code>	Business event	Funnel tracking
Deployment	<code>deployment</code>	Software development lifecycle event (same stage)	Change tracking — SDLC events, not business events, are the documented path for deployments, builds and releases
Critical error	<code>error.critical</code>	Business event	Workflow trigger — not a problem; see below

Business events do not open problems

Both processors above are **Business event** processors, and that determines what the output can do. A business event lands in Grail, is queryable, and can trigger a workflow — but it **never raises a Davis problem** and never enters problem correlation. The `error.critical` row above is the one to watch: a critical-error business event is a perfectly good workflow trigger, and it is not an incident. Nothing in the Problems app will show it, and no on-call escalation built on problem triggers will see it.

That distinction decides which processor you want:

You want	Extract as	Opens a problem?	Correlation key applies?
Analytics, funnels, KPIs, a workflow hook	Business event (<code>bizevents</code>)	No	No — not part of this path
An alert that participates in incident response	Davis event	Yes	Yes — <code>dt.smartscape_source.id</code> is mandatory

If you need the second, the Davis-event extraction path has one hard prerequisite: the event must be attributable to a Smartscape entity. Events sharing a `dt.smartscape_source.id` merge into a single problem; an event that leaves it unset is attributed to the environment entity instead, so **every** extraction across the tenant names that same entity and collapses into one problem — which, for a per-record processor firing at log volume, produces a permanently-open problem that names nothing you can act on. Since extraction can only read fields already on the record, that means the source stream must be entity-enriched first (§4 above, and the OneAgent attribute-enrichment note in this notebook's prerequisites). OPMIG-07 covers the Davis-event extraction configuration and carries a coverage query for checking the source stream; AIOPS-03 §1 covers the correlation rules.

Sources: [Avoid overalerting \(DT docs\)](#), [Data extraction stage \(DT docs\)](#) — “The Business event processor extracts a business event and re-ingests it.” and, for SDLC events, “used to track deployments, builds, and releases from CI/CD logs or generic events”, [DQL matcher in OpenPipeline \(DT docs\)](#). **Derived:** the two-row processor-choice table maps the documented correlation requirement onto the bizevents-vs-Davis-event split this section configures.


```
// Discover patterns suitable for event generation
fetch logs, from: now() - 1h
| filter contains(content, "completed")
      OR contains(content, "success")
      OR contains(content, "failed")
      OR contains(content, "created")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| summarize {count = count()}, by: {content_preview}
| sort count desc
| limit 20
```

```
// Check existing bizevents (if any generated)
fetch bizevents, from: now() - 24h
| summarize {count = count()}, by: {event.type}
| sort count desc
| limit 15
```

```
// Preview: Logs that would become critical error events
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| filter contains(content, "critical")
      OR contains(content, "fatal")
      OR contains(content, "severe")
| fieldsAdd content_preview = substring(content, from: 0, to: 100)
| fields timestamp, k8s.namespace.name, content_preview
| sort timestamp desc
| limit 20
```

6. Bucket Routing

Route logs to **appropriate buckets** based on content, source, or computed attributes.

Why Bucket Routing Matters

Bucket	Logs	Retention	Cost Impact
default_logs	Standard	35 days	Baseline
debug_logs	DEBUG/TRACE	7 days	80% savings
audit_logs	Security/compliance	365 days	Compliance
error_logs	Errors only	90 days	Investigation

OpenPipeline Route Processor

```
# Route logs to appropriate buckets
processors:
  - name: route-debug-logs
    type: route
    condition: loglevel == "DEBUG" OR loglevel == "TRACE"
    bucket: debug_logs

  - name: route-audit-logs
    type: route
    condition: matchesValue(content, "*audit*") OR matchesValue(content, "*security*")
    bucket: audit_logs

  - name: route-error-logs
    type: route
    condition: loglevel == "ERROR" OR loglevel == "FATAL"
    bucket: error_logs
```

```
// Current bucket distribution
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.system.bucket}
| sort count desc
```

```
// Preview routing decisions
fetch logs, from: now() - 1h
| fieldsAdd target_bucket = if(loglevel == "DEBUG" OR loglevel == "TRACE", "debug_logs",
                             else: if(loglevel == "ERROR" OR loglevel == "FATAL", "error_logs",
                             else: if(contains(content, "audit") OR contains(content, "security"),
"audit_logs",
                             else: "default_logs")))
| summarize {count = count()}, by: {target_bucket, loglevel}
| sort count desc
```

```
// Estimate storage savings from routing
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
  total_logs = count(),
  total_mb = sum(content_bytes) / 1048576.0,
  debug_logs = countIf(loglevel == "DEBUG" OR loglevel == "TRACE"),
  debug_mb = sum(if(loglevel == "DEBUG" OR loglevel == "TRACE", content_bytes, else: 0)) /
1048576.0
}
| fieldsAdd debug_pct = round((debug_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd savings_7d_vs_35d = round(debug_mb * 0.8, decimals: 2)
```

7. Filtering (Drop Record)

Reduce log volume by **dropping** low-value records at ingestion.

OpenPipeline Filter Processor

```
# Drop noisy, low-value logs
processors:
  - name: drop-health-checks
    type: filter
    condition: matchesValue(content, "**health*") AND loglevel == "INFO"
    action: drop

  - name: drop-heartbeats
    type: filter
    condition: matchesValue(content, "**heartbeat*") OR matchesValue(content, "**keepalive*")
    action: drop
```

There is no log-sampling processor. None appears in the complete processor list in *Processing in OpenPipeline*, and a matcher cannot sample — `random()` is rejected in OpenPipeline matchers. To drop DEBUG records outright, use a **Drop record** processor (`loglevel == "DEBUG"`) in the Processing stage. To keep metric extraction from DEBUG records while not storing them, use a **No storage assignment** processor in the Bucket assignment stage instead: *"The record continues through all pipeline stages, including metric extraction, and is not stored only at the end."*

Sources: [Processing in OpenPipeline \(DT docs\)](#) — *"The following table lists alphabetically all available processors in a pipeline."*, [OpenPipeline processing examples \(DT docs\)](#)

```
// Identify candidates for filtering (high volume, low value)
fetch logs, from: now() - 1h
| filter contains(content, "health")
      OR contains(content, "heartbeat")
      OR contains(content, "alive")
      OR contains(content, "ready")
| summarize {count = count()}, by: {k8s.namespace.name, loglevel}
| sort count desc
| limit 15
```

```
// Calculate potential savings from filtering health checks
fetch logs, from: now() - 24h
| summarize {
  total_logs = count(),
  health_logs = countIf(contains(content, "health") OR contains(content, "heartbeat")),
  debug_logs = countIf(loglevel == "DEBUG")
}
| fieldsAdd health_pct = round((health_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd debug_pct = round((debug_logs * 100.0) / total_logs, decimals: 1)
| fieldsAdd potential_reduction = round(((health_logs + debug_logs * 0.9) * 100.0) / total_logs, decimals: 1)
```

8. Complete Pipeline Example

Here is one complete log pipeline, written as the processors you add in **Settings > Process and contextualize > OpenPipeline > Logs > Pipelines**. OpenPipeline has no importable pipeline file format of its own; each row is a

processor, grouped by the stage it belongs to. Rows are listed in execution order, because *"The sequence of stages is fixed for all pipelines and cannot be modified."* Within a stage, processors run in the order you add them.

#	Stage	Processor	Matching condition	Configuration
1	Processing	Drop record	<code>matchesValue(content, "*health*") AND loglevel == "INFO"</code>	—
2	Processing	DQL	<code>matchesValue(content, "*HTTP*")</code>	<code>parse content, "LD:method SPACE '/' LD:path SPACE INT:http.status_code SPACE INT:duration_ms" — parse into http.status_code, not status, which is the log record's own status field</code>
3	Processing	DQL	<code>true</code>	<code>fieldsAdd environment = if(contains(k8s.namespace.name, "prod"), "production", else: "non-prod")</code>
4	Processing	DQL (masking — the pattern is DPL, not regex)	<code>true</code>	<code>fieldsAdd content = replacePattern(content, "[A-Za-z0-9._%+-]+ '@' [A-Za-z0-9.-]+", "[EMAIL-MASKED]")</code>
5	Bucket assignment	Bucket assignment	<code>loglevel == "DEBUG"</code>	Bucket: <code>debug_logs</code>
6	Metric extraction	Counter metric	<code>isNotNull(http.status_code)</code>	Metric key <code>log.http.requests</code> ; dimensions <code>k8s.namespace.name</code> , <code>http.status_code</code>
7	Data extraction	Business event	<code>status == "ERROR"</code>	Event type <code>com.app.error</code> (static); event provider <code>app-logs</code> (static)

What the stage order means for this pipeline:

- **Drop record (1) removes the record for good.** *"Drops a record. The record isn't processed further and isn't stored."* Health checks it drops are not counted by the metric in row 6. To count them but not store them, drop nothing and use a **No storage assignment** processor in the Bucket assignment stage instead (§7).
- **Bucket assignment (5) runs before extraction (6, 7).** Records are extracted whichever bucket they go to.
- **`contains()` is fine inside a DQL processor (row 3) and rejected in a matching condition.** The processor statement verifies as valid, while the same function in a matcher fails with the verifier error `The function contains() isn't enabled.` Every matching condition above uses `matchesValue` / `isNotNull` / `==` and verifies as a valid logs matcher.
- **Routing is not bucket assignment.** A *route* sends records to a **pipeline**. The bucket is chosen inside that pipeline, by row 5.

Sources:

~ [Processing in OpenPipeline \(DT docs\)](#) — *"The sequence of stages is fixed for all pipelines and cannot be modified."* and *"Drops a record. The record isn't processed further and isn't stored."*

~ [Metric extraction stage \(DT docs\)](#) — *"The Counter metric processor increments a counter by 1 for each matching record."*

- [Data extraction stage \(DT docs\)](#)
- [DQL matcher in OpenPipeline \(DT docs\)](#)

```
// Verify current pipeline processing
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    with_pipeline = countIf(isNotNull(dt.openpipeline.pipelines)),
    unique_pipelines = countDistinct(dt.openpipeline.pipelines)
}, by: {dt.openpipeline.source}
| sort total_logs desc
```

```
// Pipeline processing summary
fetch logs, from: now() - 1h
| summarize {count = count()}, by: {dt.openpipeline.pipelines, dt.system.bucket}
| sort count desc
```



Summary

In this notebook, you learned:

- ✓ **Parsing** - Extract structured fields at ingestion
- ✓ **Metrics** - Create dimensional metrics from logs
- ✓ **Attributes** - Add computed fields for analysis
- ✓ **Events** - Generate business events from patterns
- ✓ **Routing** - Direct logs to appropriate buckets
- ✓ **Filtering** - Drop low-value logs (or skip storage with No storage assignment)

Key Takeaway

Process at ingestion, not at query time. OpenPipeline processing is fundamental to cost optimization, data quality, and operational efficiency.



Next Steps

Continue to **OPLOGS-04: Buckets & Data Governance** to learn about storage management and retention policies.



References

- [OpenPipeline \(DT docs\)](#)

- [Processing in OpenPipeline \(DT docs\)](#)
- [Parse log lines and extract a metric \(DT docs\)](#)
- [Get business events from logs and spans \(DT docs\)](#)
- [Dynatrace Pattern Language \(DT docs\)](#)
- [OpenPipeline processing examples \(DT docs\)](#)

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